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The brain and its main anatomical subdivisions in living hominoids using magnetic resonance imaging

Primary comparative data on the hominoid brain are scarce and major neuroanatomical differences between humans and apes have not yet been described satisfactorily, even at the gross level. Basic questions that involve the evolution of the human brain cannot be addressed adequately unless the brains of all extant hominoid species are analyzed. Contrary to the scarcity of original data, there is a rich literature on the topic of human brain evolution and several debates exist on the size of particular sectors of the brain, e.g., the frontal lobe.

In this study we applied a non-invasive imaging technique (magnetic resonance) on living human, great ape and lesser ape subjects in order to investigate the overall size of the hominoid brain. The images were reconstructed in three dimensions and volumetric estimates were obtained for the brain and its main anatomical sectors, including the frontal and temporal lobes, the insula, the parieto-occipital sector and the cerebellum.

A remarkable homogeneity is present in the relative size of many of the large sectors of the hominoid brain, but interspecific and intra-specific variation exists in certain parts of the brain. The human cerebellum is smaller than expected for an ape brain of human size. It is suggested that the cerebellum increased less than the cerebrum after the split of the human lineage from the African ancestral hominoid stock. In contrast, humans have a slightly larger temporal lobe and insula than expected, but differences are not statistically significant.

Humans do not have a larger frontal lobe than expected for an ape brain of human size and gibbons have a relatively smaller frontal lobe than the rest of the hominoids. Given the fact that the frontal lobe in humans and great apes has similar relative size, it is parsimonious to suggest that the relative size of the whole of the frontal lobe has not changed significantly during hominid evolution in the Plio-Pleistocene.

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Introduction

Contrary to expectation, primary comparative data on the primate brain are scarce and this is true in particular for the brain of the apes. Lack of comparative neuroanatomical evidence from the extant hominoids constrains our understanding of the evolution of the human brain and consequently of the human mind. A large amount of information

is available on the brain of a few, select monkey species, in particular *Macaca mulatta*, the rhesus monkey. A considerable body of information is growing on the human brain as well. Nevertheless, very little is known about the brains of most other primates including the apes, whose nervous systems remain for the most part uncharted territory (Tuttle, 1986). The few comparative studies of structures of the human and

ape brains (Stephan *et al.*, 1981; Blinkov & Glezer, 1968) have suffered from incomplete representation of several ape species, effects of shrinkage on postmortem material, and from small sample sizes.

How different is the human brain from the brain of our closest relatives, the apes? Are there anatomical features in the organ of the mind, the brain, that can be identified as uniquely human? Which are the features that are shared among hominoids and which are the ones that are species specific? Regardless of how vigorously we investigate the brain of the macaque and other selected primates, we will never be able to address issues on the evolution of the brain based only on such information. An examination of the organization of the brain of many closely related extant species is imperative.

What is the size of the major sectors of the hominoid brain? Comparative information on even large anatomical sectors is not available. In a previous investigation (Semendeferi *et al.*, 1997) we began applying neuroimaging techniques in order to analyze the size of the frontal lobes in a small sample of postmortem brain specimens of apes and humans. Here we report results from a larger group of ape and human brains, studied *in vivo* with magnetic resonance imaging (MRI). The sectors reported here include the whole brain, the hemispheres, the frontal, temporal, insular, and parieto-occipital lobes, and the cerebellum.

Materials and methods

Materials

In this study we decided to scan the brain of living representatives of all extant hominoid: humans, African great apes (bonobos, chimpanzees and gorillas), Asian great apes (orang-utans) and Asian lesser apes (gibbons). The brains of ten normal human subjects (from the Department of Neurology, University of Iowa) and of 19 ape subjects (from the Yerkes Regional

Primate Research Center) were scanned using thin (1.2–1.6 mm), contiguous, T1 weighted MR images. The subjects include adult individuals of both sexes. The non-human hominoids include six chimpanzees, three bonobos, two gorillas, four orang-utans, and four gibbons (Figure 1).

Methods

Thin cut MR images were obtained for all human subjects in a General Electric 4096 Plus scanner operating at 1.5 Tesla, using the following protocol: SPGR/50, TR 24, TE 5, NEX 2, FOV 24 cm, matrix 256 × 192, 124 contiguous coronal slices, 1.6 mm thick. The apes were scanned in a Phillips Gyroscan NT and thickness of axial slices varied between 1.2 and 1.4 mm with a slice overlap of 0.6 mm (FOV 124–230 cm).

All brains were reconstructed in three-dimensions using Brainvox (Damasio & Frank, 1992; Frank *et al.*, 1997), an interactive family of programs designed to reconstruct, segment and measure brains from MR acquired images. All segmentations and measurements were achieved using Silicon Graphics computer stations in the following manner.

Hemispheres. After each brain was segmented from all non-brain tissue and reconstructed in three-dimensions it was traced to exclude the following: pons, medulla and the greater part of the midbrain. On each coronal slice where these structures are visible, we connected the two transverse fissures with a straight line. This eliminated all structures below the line (those mentioned above). Then, the two hemispheres were separated. Within the hemispheres, the frontal, temporal, parieto-occipital and insular regions were manually segmented using the following guidelines:

Frontal lobe. On the lateral surface the posterior limit of the frontal lobe was defined by the central sulcus. Whenever the central sulcus did not reach the Sylvian

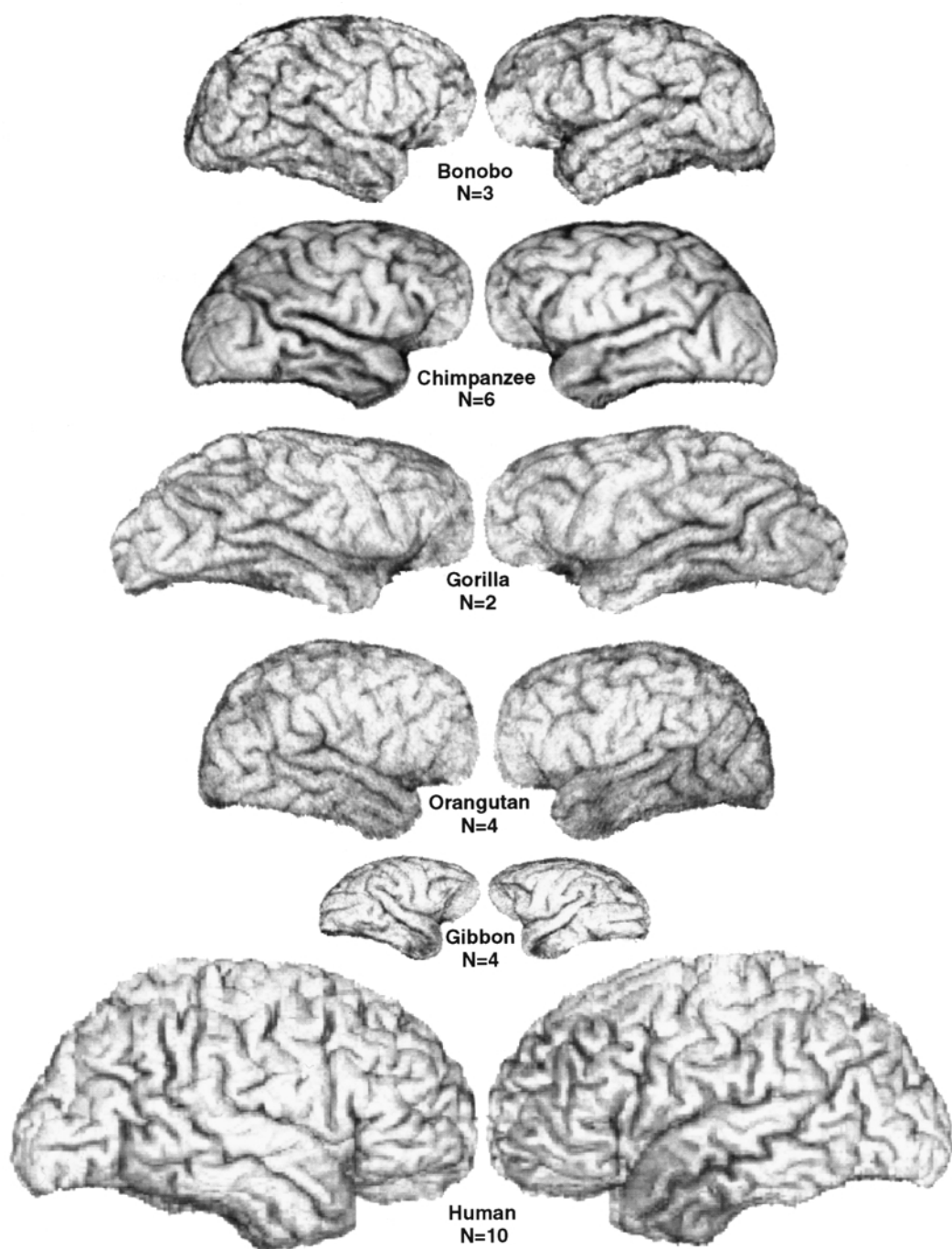


Figure 1. Three-dimensional reconstruction of the brain of living representatives of all extant hominoids.

(a)

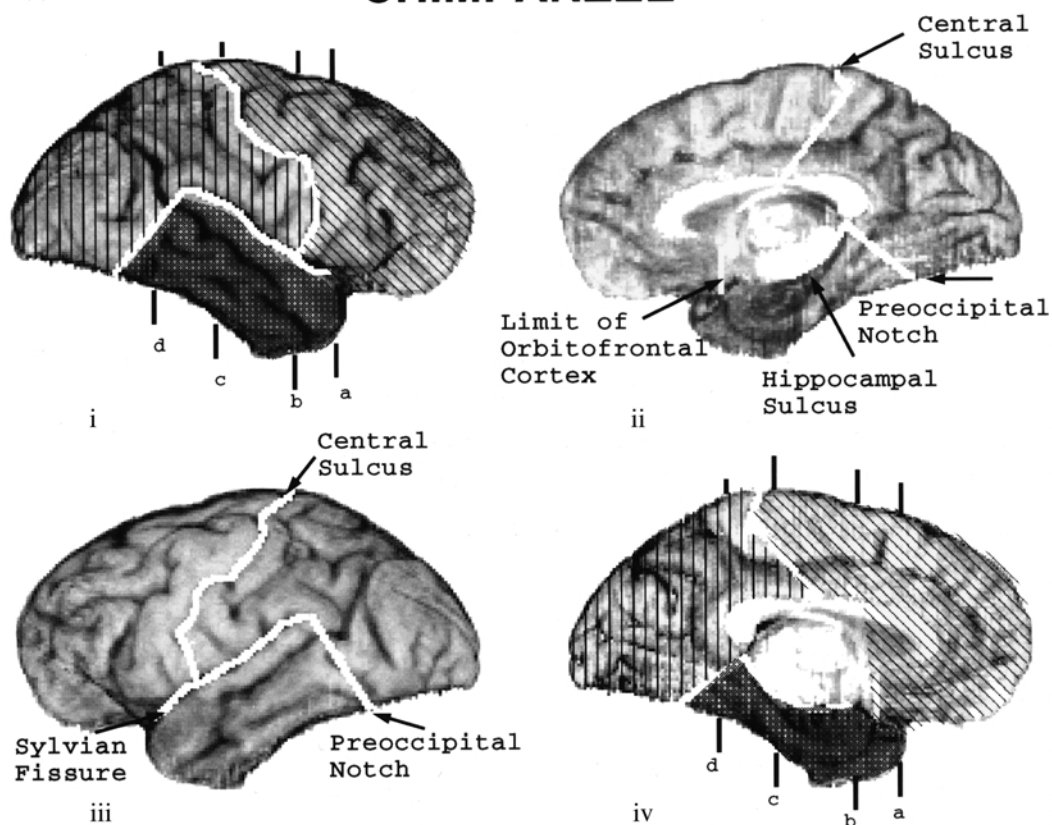
CHIMPANZEE

Figure 2 (a) (legend on facing page).

fissure we prolonged it so as to intersect this fissure. On the mesial surface the end of the central sulcus was connected to the end of the orbital sector. All structures located rostrally were included in this sector (cortex, white matter, part of the ventricles and the underlying subcortical structures) [Figure 2(a)].

Temporal lobe. The posterior limit of the temporal lobe, on the lateral surface, was defined by a straight line connecting the end of the Sylvian fissure with the preoccipital notch. On the mesial surface the preoccipital notch was connected with the caudal end of the hippocampal sulcus thus providing the posterior limit of the temporal lobe on the

mesial aspect of the hemisphere (cortex, white matter, part of the ventricles and subcortical structures underlying the temporal cortices were included).

Insula. The insula was identified in each coronal section and separated from the surrounding cortices using the circular insular sulcus and the extreme capsule as limits [Figure 2(b)].

Parieto-occipital region. We decided, at this time, not to separate the occipital from the parietal region because there is no consistent landmark, across species, that could be used to separate these two sectors on the dorsolateral aspect of the hemispheres.

(b)

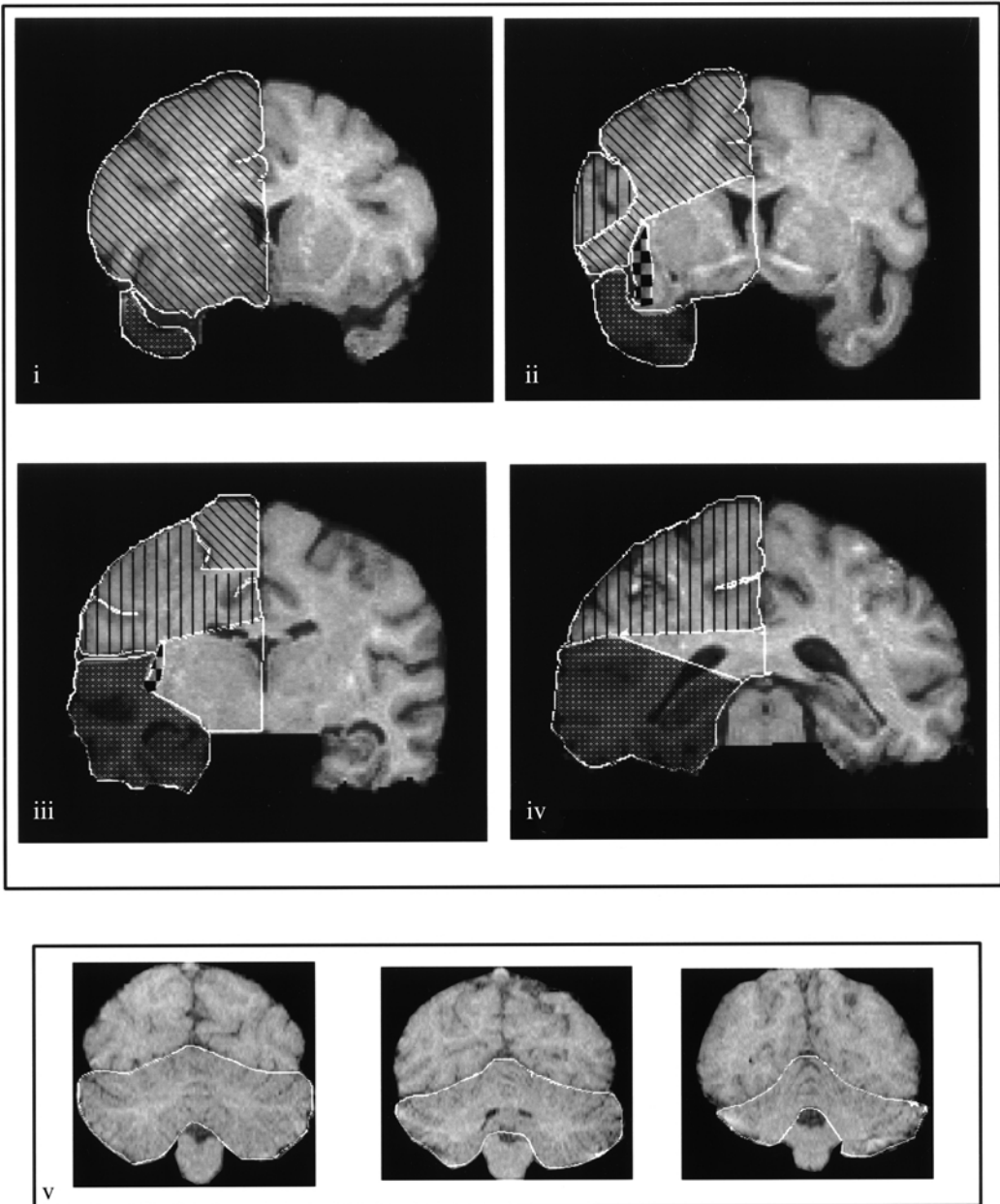


Figure 2 (b).

Figure 2. (a) Three-dimensional reconstruction of the right and left hemispheres (lateral and mesial views) of a chimpanzee brain. The right hemisphere and its main subdivisions are shown in the upper left (lateral view) and upper right (mesial view). The same subdivisions (frontal, temporal, parieto-occipital and core) are shown also on the left hemisphere (lateral and mesial views) at the lower part of the figure. Vertical bars i, ii, iii, and iv indicate plans of cut for the coronal sections shown in the next figure. Vertical parallel strips highlight the frontal lobe, tilted parallel strips highlight the parieto-occipital sector and dots highlight the temporal lobe. (b) Coronal sections through the brain of the chimpanzee at various levels starting rostrally (i) and continuing caudally (iv) towards the posterior end of the brain. The frontal, temporal, parieto-occipital, insular and core regions of interest are shown. In the lower part of the graph (v) is an example of the segmentation of the cerebellum from the rest of the brain in one of the gorilla subjects. Areas in the coronal sections are marked as in [Figure 2\(a\)](#). Insula is highlighted with large black squares.

The anterior limit of this sector in relation to the frontal lobe is provided by the central sulcus, while the posterior limits of the temporal lobe, as discussed above, provide the remaining anterior borders of this sector on the lateral and mesial surfaces. This sector includes all parietal and occipital cortices and underlying white matter.

The hemispheres are thus composed of the summation of the frontal, temporal, parieto-occipital and insular regions of interest (rois) plus the central subcortical grey and white matter not included in the above sectors. Specifically, due to their topographic location underneath the frontal and temporal cortices, the most rostral part of the basal ganglia and the amygdala were included in the frontal and temporal rois respectively. This way consistency in the delineation of the lobes across species was guaranteed. The central or "core" structures refer to all other subcortical structures of the hemispheres except for the above. Although delineation and measurement of individual subcortical structures is highly desirable, such an attempt requires different segmentation procedures and will therefore be addressed separately in future research.

Cerebellum. The cerebellum was identified as a separate roi in each coronal section by exclusion of the hemispheres and the brainstem [Figure 2(b)].

Brain. The brain roi includes the two hemispheres plus the cerebellum.

Results

Absolute values

Results with mean values per species as well as individual values are presented in Tables 1 and 2. The absolute values of the brain and its components form three plateaus: humans have the largest brains, followed by the great apes and then the small bodied

gibbons (Table 1, Figure 3). The distribution of the values among the great apes varies depending on the sector investigated.

In the case of the whole brain (two hemispheres plus cerebellum), orang-utans have the largest mean value, followed closely by the gorillas, the bonobos and the chimpanzees [Table 1, Figure 3(a)]. With respect to the size of the cerebellum it is the gorillas who show the largest cerebellum, followed by the orang-utans, the bonobos and the chimpanzees [Table 1, Figure 3(b)].

The two hemispheres together in humans have a mean volume of 1233.8 cm³. As expected on the basis of the values obtained from the whole brain, the orang-utans and gorillas have the largest hemispheres among the apes (391.2 cm³ and 355.9 cm³) followed by the bonobos and the chimpanzees (290.5 cm³ and 279.7 cm³) [Table 1, Figure 3(c)].

The frontal lobes form a large segment of the hemispheres in all hominoids and as is the case with the volume of the two hemispheres, it is the orang-utans among the apes who have the largest frontal lobe, followed by the gorilla, bonobo and chimpanzee [Figure 3(d)]. Orang-utans and gorillas also have the largest mean values among apes for the temporal lobes [Figure 3(e)]. The insula is largest in the gorilla and orang-utan among the apes [Figure 3(f)] and this is also the case for the parieto-occipital sector [Figure 3(g)].

Relative values

The results for this section, mean values per species and values for each individual subject can be found in Tables 3 and 4. The frontal lobes make up 36.8% of the human hemispheres (35–38.5%), 34.9% in the bonobo (33.8–37%), 34.8% in the chimpanzee (33.3–36.2%), 35.1% in the gorilla (34.9 and 35.3%), 36.3% in the orang-utan (35.3–37.1%), and 28.4% in the gibbon (27.2–29.6%). Humans and orang-utans have the largest frontal lobes in relative

Table 1 Mean absolute volume of the brain and its major subdivisions*

	Whole Brain†		Cerebellum		Hemispheres‡		Frontal§		Temporal§		Hemispheres‡		Parieto-occipital§		Core§	
	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.
Human	1390.3	35.0	155.1	3.5	1233.8	32.9	454.8	14.9	223.2	7.7	17.4	0.6	448.8	13.7	89.9	2.1
Bonobo	336.3	11.2	45.8	2.3	290.5	9.6	101.5	3.8	48.2	2.5	3.7	0.2	109.1	3.5	28.0	1.4
Chimpanzee	320.9	27.9	41.2	3.3	279.7	24.8	97.2	8.3	48.2	4.2	3.1	0.2	103.6	9.8	27.7	2.7
Gorilla	425.1	66.5	69.3	15.7	355.9	50.9	124.8	17.2	49.8	7.3	5.3	1.8	142.6	24.0	33.5	0.7
Orang-utan	443.2	28.3	52.0	3.4	391.2	26.5	142.3	11.1	65.7	5.1	4.8	0.5	138.8	9.7	39.6	2.3
Gibbon	82.3	3.5	10.7	1.0	71.6	2.6	20.3	1.1	13.5	0.5	0.7	0.0	28.1	0.5	9.0	0.6

*Values are in cm³; †whole brain includes cerebellum plus the two hemispheres; ‡both hemispheres are included; §values include sectors from both hemispheres.

Table 2 Individual values of the absolute volume of the brain and major subdivisions*

	Whole brain	Cerebellum	Hemispheres†	Frontal	Temporal	Insula	Parieto-occipital	Core
Human 1	1369.5	152.6	1216.9	440.2	210.0	16.6	460.6	89.5
Human 2	1381.5	149.3	1218.4	462.1	229.6	18.8	418.4	89.5
Human 3	1231.1	144.4	1086.7	397.0	215.9	14.6	379.0	82.9
Human 4	1274.9	159.0	1115.9	399.0	211.7	17.7	392.4	95.1
Human 5	1489.6	167.9	1321.7	508.5	252.5	19.0	444.0	97.7
Human 6	1477.9	159.5	1318.4	461.3	232.9	18.0	510.4	95.8
Human 7	1608.4	171.2	1437.2	551.1	261.5	16.9	508.9	98.8
Human 8	1372.0	133.4	1238.6	461.4	234.6	16.5	447.2	78.9
Human 9	1315.0	152.4	1162.6	423.4	176.4	15.2	463.7	83.9
Human 10	1382.9	161.2	1221.7	443.8	207.1	20.3	463.3	87.2
Bonobo 1	333.6	48.5	285.1	105.6	44.9	3.5	104.7	26.4
Bonobo 2	356.9	47.7	309.2	105.0	53.2	4.1	116.1	30.8
Bonobo 3	318.3	41.1	277.2	93.8	46.6	3.4	106.6	26.8
Chimpanzee 1	322.1	41.9	280.2	99.7	47.1	2.6	99.0	31.8
Chimpanzee 2	242.4	34.9	207.5	74.1	35.3	2.5	76.5	19.1
Chimpanzee 3	269.8	34.9	234.9	85.0	38.9	2.7	85.7	22.6
Chimpanzee 4	304.9	35.1	269.8	89.8	51.1	3.5	101.3	24.1
Chimpanzee 5	437.5	55.3	382.2	133.4	64.3	4.0	144.3	36.2
Chimpanzee 6	348.6	45.2	303.4	101.2	52.2	3.0	114.5	32.5
Gorilla 1	358.6	53.6	305.0	107.6	42.5	3.5	118.6	32.8
Gorilla 2	491.6	84.9	406.7	142.0	57.0	7.1	166.5	34.1
Orang-utan 1	456.0	61.0	395.0	142.0	63.2	5.9	139.2	44.7
Orang-utan 2	508.5	52.7	455.8	169.3	74.8	4.3	165.1	42.3
Orang-utan 3	436.9	49.2	387.7	142.8	72.3	5.3	132.0	35.3
Orang-utan 4	371.3	45.2	326.1	115.1	52.4	3.7	118.8	36.1
Gibbon 1	79.5	8.5	71.0	19.5	13.7	0.8	28.0	9.0
Gibbon 2	92.6	13.5	79.1	23.4	14.9	0.7	29.6	10.5
Gibbon 3	78.5	10.0	68.5	18.6	13.0	0.7	27.3	8.9
Gibbon 4	78.5	10.7	67.8	19.8	12.5	0.6	27.3	7.6

*Values are in cm³; †both hemispheres are included.

terms, while gibbons have the smallest [Table 3, Figure 4(a)]. Human relative values for the frontal lobe overlap with the range of values of all great apes, in particular with those of the orang-utan, while gorilla values fall at the lowest end of the relative values of the human range. Gibbons are the only hominoid that do not have overlapping values with any of the great apes, including humans and differ from them by 4% to more than 10%.

The temporal lobes in humans make up 18.1% of the hemispheres (15.2–19.9%), bonobos 16.6% (15.7–17.2%), chimpanzees 17.2% (16.6–18.9%), gorillas 14% (13.9 and 14%), orang-utans 16.8% (16–18.6%),

and gibbons 18.9% (18.4–19.3%). The range of relative values for the human brain includes the ranges of all other great and lesser apes with the exclusion of the two gorillas whose relative size of this sector is close to the lowest end of the variation [Table 3, Figure 4(b)].

The insula in humans is 1.4% (1.2–1.7%), bonobos 1.3% (1.2–1.3%), chimpanzees 1.1% (0.9–1.3%), gorillas 1.4% (1.1 and 1.7%), orang-utans 1.2% (1–1.5%), and gibbons 1% (0.9–1.1%). The range of values of the human insula overlaps with the range of values of all other hominoids, most of which are around the lower end of the human variation [Table 3, Figure 4(c)].

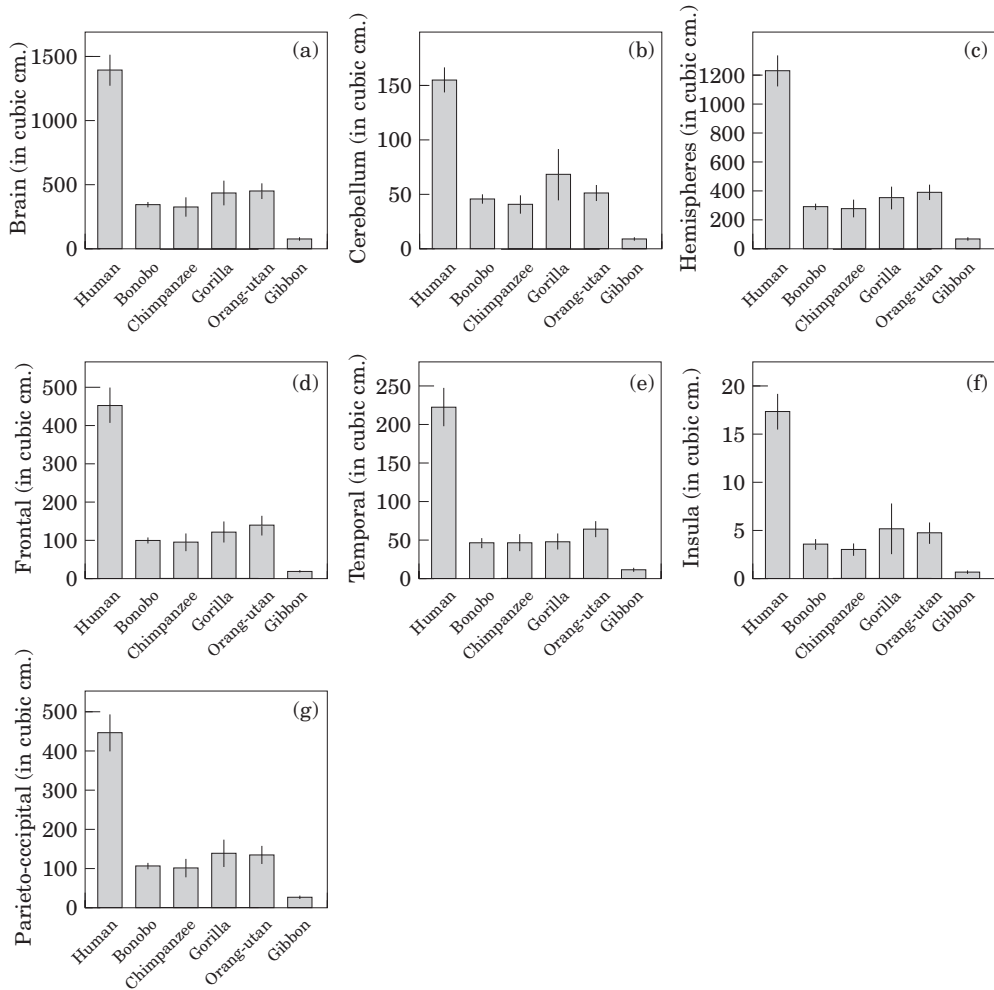


Figure 3. Graphic representation of the absolute volume and standard deviation of the brain (a), cerebellum (b), two hemispheres (c), frontal lobe (d), temporal lobe (e), insula (f), and parieto-occipital region (g). On the x-axis are the names of the species and on the y-axis the volume of the sectors in cm³ (bars represent standard deviation).

The parieto-occipital region is 36.4% in humans (33.6–39.9%), bonobos 37.6% (36.7–38.5%), chimpanzees 37% (35.3–37.8%), gorillas 39.9% (38.9–41%), orang-utans 35.5% (34.1–36.4%), and gibbons 39.2% (37.4–40.3%). The range of relative values for the parieto-occipital region of the human brain includes most of the values of the non-human hominoids, with some of the gibbons and gorillas being close to the

highest end of the range [Table 3, Figure 4(d)].

As a percentage of the brain, the cerebellum in humans is 11.2% (9.7–15.5%), bonobos 13.6% (12.9–14.5%), chimpanzees 12.9% (11.5–14.4%), gorillas 16.1% (14.9 and 17.3%), orang-utans 11.8% (10.4–13.4%), and gibbons 12.9% (10.7–14.6%) (Table 3). Gorillas have the largest mean relative value for the cerebellum, while

Table 3 Mean relative values of the subdivisions of the hominoid brain (as percentage of the hemispheres)

	Cerebellum*		Frontal		Temporal		Insula		Parieto-occipital		Core	
	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.	Mean	S.E.
Human	11.2	0.2	36.8	0.4	18.1	0.4	1.4	0.0	36.4	0.7	7.3	0.2
Bonobo	13.6	0.5	34.9	1.0	16.6	0.4	1.3	0.0	37.6	0.5	9.6	0.2
Chimpanzee	12.9	0.4	34.8	0.5	17.2	0.4	1.1	0.1	37.0	0.4	9.9	0.4
Gorilla	16.1	1.2	35.1	0.2	14.0	0.0	1.4	0.3	39.9	1.0	9.6	1.2
Orang-utan	11.8	0.6	36.3	0.4	16.8	0.6	1.2	0.1	35.5	0.5	10.2	0.6
Gibbon	12.9	0.8	28.4	0.6	18.9	0.2	1.0	0.1	39.2	0.6	12.5	0.5

*As percentage of the brain.

Table 4 Individual values of the relative volume of the major subdivisions of the hominoid brain*

	Frontal	Temporal	Insula	Parieto-occipital	Core	Cerebellum*
Human 1	36.2	17.3	1.4	37.9	7.6	11.1
Human 2	37.9	18.8	1.5	34.3	7.5	10.8
Human 3	36.5	19.9	1.3	34.9	7.4	11.7
Human 4	35.8	19.0	1.6	35.2	8.5	15.5
Human 5	38.5	19.1	1.4	33.6	7.4	11.3
Human 6	35.0	17.7	1.4	38.7	7.3	10.8
Human 7	38.3	18.2	1.2	35.4	6.9	10.6
Human 8	37.3	18.9	1.3	36.1	6.4	9.7
Human 9	36.4	15.2	1.3	39.9	7.2	11.6
Human 10	36.3	17.0	1.7	37.9	7.1	11.7
Bonobo 1	37.0	15.7	1.2	36.7	9.4	14.5
Bonobo 2	34.0	17.2	1.3	37.5	10.0	13.4
Bonobo 3	33.8	16.8	1.2	38.5	9.7	12.9
Chimpanzee 1	35.6	16.8	0.9	35.3	11.4	13.0
Chimpanzee 2	35.7	17.0	1.2	36.9	9.2	14.4
Chimpanzee 3	36.2	16.6	1.1	36.5	9.6	12.9
Chimpanzee 4	33.3	18.9	1.3	37.6	8.9	11.5
Chimpanzee 5	34.9	16.8	1.0	37.8	9.5	12.6
Chimpanzee 6	33.4	17.2	1.0	37.7	10.7	13.0
Gorilla 1	35.3	13.9	1.1	38.9	10.8	14.9
Gorilla 2	34.9	14.0	1.7	41.0	8.4	17.3
Orang-utan 1	36.0	16.0	1.5	35.2	11.3	13.4
Orang-utan 2	37.1	16.4	1.0	36.2	9.3	10.4
Orang-utan 3	36.8	18.6	1.4	34.1	9.1	11.3
Orang-utan 4	35.3	16.1	1.1	36.4	11.1	12.2
Gibbon 1	27.5	19.3	1.1	39.4	12.7	10.7
Gibbon 2	29.6	18.8	0.9	37.4	13.3	14.6
Gibbon 3	27.2	19.0	1.0	39.9	13.0	12.7
Gibbon 4	29.2	18.4	0.9	40.3	11.2	13.6

*As percentage of brain (=two hemispheres+cerebellum).

orang-utans have the smallest mean value of all apes including the gibbons. Humans have a smaller cerebellum when compared

to the rest of the hominoids, but the range of relative values is overlapping across hominoids [Table 3, Figure 4(e)].

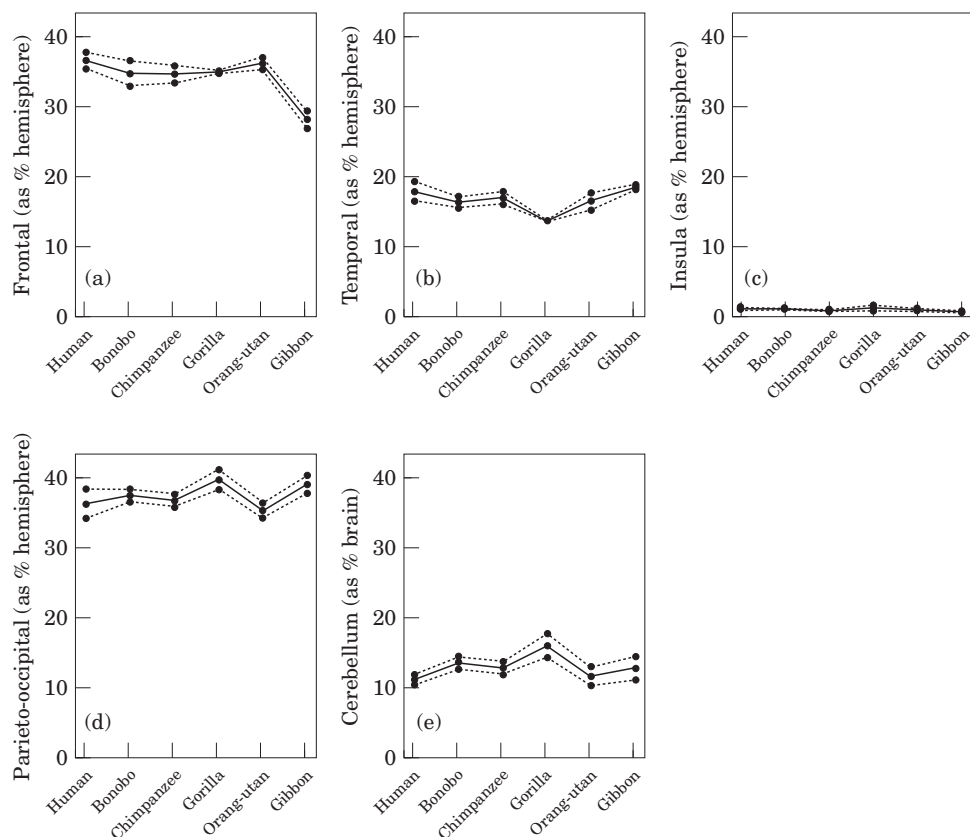


Figure 4. Graphic representation of the relative values of the frontal lobe (a), temporal lobe (b), insular (c), parieto-occipital region (d), and cerebellum (e).

Allometry

Values for the above sectors were regressed against brain size and hemisphere size using least square (LS) (Smith, 1994) and reduced major axis (RMA) (Aiello, 1992). We fit a restricted-range linear regression (Figure 5) through the apes only, and another one through humans. We then performed a significance test for the slopes and intercepts of the two lines being different from each other. Under the null hypothesis of no difference, the lines should be approximately coincident.

The frontal lobes of the human brain are as large as expected for an ape brain of human size [Figure 5(a)] and differences in the slopes are not significant ($b=1.34$,

$P=0.18$ with LS; $b=1.75$, $P=0.08$ with RMA). By visual inspection the temporal lobes of the human brain seem to be somewhat larger than expected for an ape brain of human size [Figure 5(b)], but differences are not significant ($b=0.38$, $P=0.7$ with LS; $b=1.44$, $P=0.15$ with RMA). The human insula appears also somewhat larger than expected for an ape brain of human size [Figure 5(c)], but again differences are not significant ($b=1.28$; $P=0.2$ with LS; $b=0.24$, $P=0.81$ with RMA) particularly due to the variation in the human group. In the case of the parieto-occipital, human values are as large as expected for an ape of human size [Figure 5(d)] with non-significant differences in the slope ($b=0.31$,

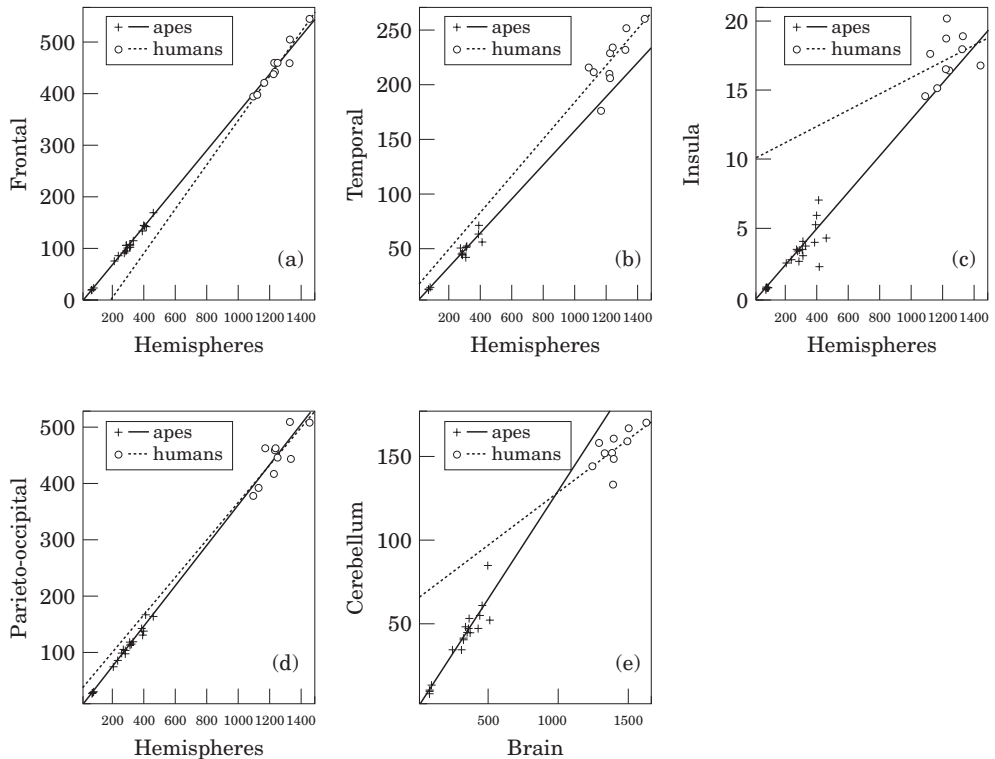


Figure 5. Allometric plots of the volume of the frontal lobe (a), temporal lobe (b), insular (c), parieto-occipital region (d), and cerebellum (e). The best fit lines are estimated on the basis of the nonhuman primates (solid lines) and humans (dotted lines).

$P=0.76$ with LS; $b=0.51$, $P=0.61$ with RMA). The cerebellum in humans is significantly smaller than expected for an ape brain of human size when LS is applied ($b=2.2$, $P=0.025$), although the RMA analysis does not show statistical significance.

Discussion

Specific findings of the present study

Great apes and humans have overlapping relative values for the size of the frontal lobe, in keeping with results presented in our previous study (Semendeferi *et al.*, 1997) based on postmortem specimens. The size of the frontal lobe, as a whole, does not distinguish the human brain among the hominoids nor is it larger than expected for an ape brain of human size.

Our earlier finding had to be tentative due to the small number of specimens, however with the present study we feel confident that the finding in fact contradicts the popular idea that humans have a disproportionately big frontal lobe (Deacon, 1988). This statement does not exclude, however, the possibility that the difference between the human frontal lobe and the frontal lobe of the great apes may reside in a relatively larger prefrontal sector. However, because there are not consistent macroscopic landmarks separating the prefrontal region from the premotor region, we elected to leave this question unanswered at this time. If reliable separation can be devised for all species involved, it would be of major interest to address this additional question.

On the other hand the relative size of the cerebellum, as a percentage of total brain volume, is smaller in humans than in other hominoids. Furthermore humans have a significantly smaller cerebellum than expected for an ape brain of human size.

Contrary to expectation, it was not in the most arboreal species that we found the largest cerebellum, a structure known to be heavily involved in fine motor tuning and balance. It was actually in the most terrestrial species among the apes, the gorilla, where we found the largest relative size for the cerebellum. However, we consider this a preliminary finding at best. The larger mean value for this species is largely due to the large cerebellum of one of the two individuals examined. This gorilla might be an outlier. Nevertheless in a previous study, based again on one individual from this species, [Stephan *et al.* \(1981\)](#) also found a similarly large cerebellum. So we could question which of the two subjects in our present study is in fact the outlier. For gorillas we found a larger parieto-occipital region and a smaller temporal lobe than in the other great apes, but our findings should be viewed as preliminary until a larger sample of gorillas can be examined. The relative size of the cerebellum, frontal lobe, temporal lobe, insula, and parieto-occipital region are similar in the bonobo, chimpanzee and orang-utan.

A remarkable homogeneity is present in the relative size of many of the large sectors of the hominoid brain, a finding that supports previous work on the size of many sectors of the primate brain ([Finlay & Darlington, 1995](#)). Nevertheless, inter-specific and intraspecific variation exists in certain parts of the brain. Gibbons have uniquely small frontal lobes when compared to the rest of the hominoids. In contrast, the relative size of other sectors of their brain (as for instance the temporal lobe or parieto-occipital sector) fall within the range of variation of the great apes and humans.

These findings underscore once more that the use of only one representative from each species is inadequate for appreciating species-specific patterns. Previous reports that discuss the evolution of the human brain based on one or two hemispheres of selected primate species might not have addressed the issues adequately and here may in fact lie the explanation of some of the discrepancies we are now finding.

Comparison of our results with previous studies

Few comparative studies have addressed the issue of differences in the size of various sectors in the hominoid brain ([Brodmann, 1912](#); [Leboucq, 1928](#); [Tilney, 1928](#); [Weil, 1929](#); [Blinkov & Glezer, 1968](#); [Stephan *et al.*, 1981](#); [Matano *et al.*, 1985](#); [Zilles *et al.*, 1988](#); [Uylings & van Eden, 1990](#)) and most of them have dealt mainly with the size of cortical surface and not the volume of various sectors. [Stephan *et al.* \(1981\)](#) measured the size of various structures in the brain of primates and other mammalian species. They included one individual from each of the following hominoids: human, chimpanzee, gorilla and gibbon. The values they reported for total brain weight were 1330 g (human), 405 g (chimpanzee), 500 g (gorilla) and 102 g (gibbon), and for the volume of the cerebellum their finding was 137.4 cm³ (human), 43.6 cm³ (chimpanzee), 69.2 cm³ (gorilla), and 12.1 cm³ (gibbon). These values fall within the range of the volumes reported here, although the gorilla and gibbon brain weights are at the high end of the range in our sample. [Stephan *et al.* \(1981\)](#) did not mention any values for the size of individual lobes.

[Blinkov & Glezer \(1968\)](#) on the basis of a large sample (over 1000 individuals) report the weight of the human cerebrum and cerebellum as 1130–1369 g and 127–149 g respectively, values that are largely overlapping with our own findings. Several studies have dealt with the size of the entire brain and endocranial capacity of hominoids

(Bonin, 1937; Kennard & Willner, 1941; Kennedy *et al.*, 1989; Tobias, 1971; Tuttle, 1986) and some variability is present in the values reported in these studies.

The results discussed here are also similar to our own previous findings (Semendeferi *et al.*, 1997) that addressed the size of the hemispheres and the frontal lobe in humans and apes based on a few postmortem brain specimens. Both the size of the hemispheres and of the frontal lobes reported in 1997 fall largely within the range of values presented here. However for the orang-utan and the gibbon the postmortem values were smaller (for both sectors). This difference could well be due to the effect of shrinkage involved in the preservation of postmortem tissue.

Evolution of the brain during the Plio-Pleistocene

The comparative data presented here on the hominoid brain allow us to make certain suggestions regarding possible evolutionary events pertaining to the brain during the Plio-Pleistocene. The overall relative size of large sectors of the brain including the frontal lobe and the parieto-occipital region does not seem to have increased in hominids after the split of the hominid line from the rest of the hominoids. Instead, the frontal lobes might have increased in size considerably earlier than the appearance of the first human ancestors, sometime after the split of the hominoids from the line leading to extant lesser apes. Following those early increases in the relative size of the hominoid frontal lobe, no further changes seem to have taken place during the Plio-Pleistocene with regard to relative overall size. It remains to be seen if subdivisions of the frontal and parieto-occipital regions into subsectors reveal differences in their relative sizes.

Figure 5 suggests that the lobe that may have increased more than other large areas of the brain during hominid evolution could prove to be the temporal lobe (or one or more of its subdivisions). Nevertheless our

sample is not large enough to be confident about the result given that the allometric plotting fails to reach statistical significance, even though visual inspection favors such a conclusion. The same is true for the insula.

After the split from the rest of the hominoids the cerebellum increased less than the cerebrum in the hominids of the Plio-Pleistocene. At the same time, the line leading to present day gorillas exhibited a specialization by having a larger cerebellum than the rest of the hominoids.

Conclusion

The data presented here suggest a remarkable homogeneity in the relative size of many of the large sectors of the hominoid brain. Human frontal lobes are not larger than expected for an ape brain of human size. The present results obtained in a larger group of individuals support the preliminary data we obtained in our previous study on the postmortem material of one single individual per species, where, contrary to expectation, a large frontal lobe was not a uniquely human feature. The human temporal lobe looks slightly larger than expected for an ape brain of the human size, but the differences are not statistically significant. The difference between humans and the rest of the hominoids with respect to large anatomical sectors may reside in a smaller relative size of the human cerebellum.

If we are to talk about what is uniquely human in relation to the brain, we cannot rely exclusively on comparisons coming from more distant taxa like other mammals or even other anthropoids to the exclusion of the apes. A comparison between the human and gibbon data, would certainly lead to different conclusions from those we reach here regarding the evolution of the human frontal lobe (gibbons have in fact a smaller relative frontal lobe, but great apes do not). One of the reasons why there has been considerable confusion, in the past,

regarding the evolution of the human frontal lobe might have been the fact that all previous studies included one or two hemispheres of only selected primate species (mostly nonhominoid anthropoids and prosimians). This fact would have affected the interpretation of the data and lead to the erroneous conclusion in favor of a uniquely large human frontal lobe.

Specialization of certain structures as reflected in their relative size can be identified even at a gross level. On the basis of even a small sample, the brain of the gorilla seems to be more specialized than that of other great apes, with a larger cerebellum, a smaller temporal lobe, and a larger parieto-occipital region. It is premature nevertheless to start to relate these gross anatomical differences with ecological and cognitive specializations, because use of a larger sample is needed along with an investigation of smaller subsectors of the brain.

Although many of the sectors of the hominoid brain exhibit similar values in both relative and absolute terms, intraspecific and interspecific variability is present. Comparisons between distant taxa or between, for example, humans and chimpanzees only, to the exclusion of the rest of the apes, might not be sufficient to understand events regarding the evolution of the hominid brain. Identification of species-specific adaptations is certainly going to be of the outmost importance.

The finding of the frontal lobes, reported here and in our previous study, is of particular interest to studies of paleoneurology. If human and great ape frontal lobes have similar relative sizes, then it is *not* likely that the frontal lobes of any hominid will show considerable relative differences in gross size. Possible differences in the relative size of the frontal lobe of any individual hominid endocast can be rather attributed to intraspecific variation, clearly present in all hominoids. In contrast, identification of the presence of differences in the relative size of

the cerebellum among Plio-Pleistocene hominid species can be very revealing with respect to timing the onset of the differential increase between the cerebellum and the cerebrum during human evolution.

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